Abhishek Mathpati

Bachelors in Computer Science & Engineering
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Education

Program	Specialization	Institute	Year	% / CGPA
BE	Computer Science & Engineering	Keshav Memorial Engineering College	2026	8.41 / 10
12th TSBIE	Mathematics, Physics & Chemistry	Narayana Junior College	2022	96.9 %
10th SSC	-	St. Mark's Boys Town High School	2020	10 / 10

Professional Experience

- **Software Engineering Intern | Teleparadigm Networks Ltd.** (Sept – Nov 2025)
  - Developed a secure, **decentralized file-sharing** platform with role-based access controls to protect sensitive user data from **unauthorized access**. Implemented distributed storage methods and **dynamic permission controls** across network nodes, maintaining **high data privacy** while removing **single points of failure**.

- **Cybersecurity Intern | Women Safety Wing, Telangana Police** (Dec 2024 – Jan 2025)
  - Assisted law enforcement officers in active **cybercrime investigations** by analysing digital evidence and performing **digital forensic evaluations**. Conducted research on **open-source intelligence (OSINT)** tools and methodologies, leveraging intelligence gathering and digital traces to trace threat actors, **identify suspects**, and support formal investigative documentation.

Projects

- **SecureShare | Decentralized Secure File Sharing and Communication Platform** (Sep 2025 – Mar 2026)
  - Developed a **decentralized**, **tamper-evident** messaging platform using **blockchain** anchoring and **IPFS storage**, backed by a **modular FastAPI backend** and a hardened Flutter client. Engineered real-time WebSocket communication enforced by JWT authentication, strict **RBAC**, and **admin audit visibility**, while integrating ephemeral **file-sharing controls** with anti-screenshot and screen-recording protections to **eliminate data leakage** and centralized points of compromise.
- **Question Paper Automation Tool (QPAT) | Automated Academic Assessment System** (April – Aug 2025)
  - Engineered a blueprint-driven **automated question paper generation** platform using a **FastAPI backend** and **Flutter** frontend, strictly aligned with **academic regulations**. Integrated **NLP-powered semantic analysis**, Bloom's Taxonomy classification, Course Outcome mapping, and automated syntax validation, while generating password-protected PDFs alongside BT-level summary analytics for **secure, quality-controlled distribution**.
- **Social Media Investigative Assistant (SMIA) | Digital Forensic Documentation Tool** (Jan – May 2025)
  - Designed a **Flutter-based forensic investigation mobile application** enabling structured, case-oriented **social media analysis** for authorized **law enforcement**. Engineered timestamp-validated screenshot **evidence capture** with automated PDF report generation, backed by **secure local storage** and cloud-synced metadata under a compliance-first, **zero-unauthorized-scraping architecture** designed for modular multi-platform extensibility.
- **College Website - TECHTONIC '24 Hackathon Winner** (Feb – May 2024)
  - Designed a **responsive, high-performance college web portal** using HTML, CSS, and JavaScript, focusing on **optimized load speeds, cross-platform accessibility, and intuitive navigation**. **Outperformed 30+ competing teams** to **secure 1st place in a national-level hackathon** by delivering a **scalable, user-centric solution** within an accelerated development timeline.
- **Park Hub: Parking Management System** (April– Sep 2023)
  - Developed a **smart urban parking** mobile application using **Flutter** and **Firebase** to optimize space allocation and streamline driver navigation. Integrated interactive mapping with **real-time spot availability**, **dynamic pricing models**, and **secure payment** processing, **significantly reducing parking search times** and enhancing end-to-end user transit efficiency.

Publications

- **Quantifying the Security-Utility-Latency Frontier: An Empirical Benchmark of Small and Medium Language Models Under Adversarial Injection** [IEEE ISDDC, Puducherry] (Accepted) (Sep 2026)
  - Evaluated five open-weight language models (Llama-3.2, Qwen-2.5) across 2,500 test queries against direct jailbreaks, indirect prompt injection, and obfuscation attacks on an isolated testbed; characterized the multi-objective Pareto frontier across attack robustness, over-refusal rates, and tail latency.
- **An Intelligent Decentralized Secure File Sharing System with Privacy-Preserving Access Control** [IEEE IC(SEC)², Coimbatore] (Accepted and Presented: To appear in IEEE Xplore) (Aug 2026)
  - Presented at the 2nd IEEE International Conference on Smart Systems for Electrical, Electronics, Communication and Computer Engineering; designed a decentralized file-sharing architecture integrating Blockchain, IPFS, Shamir's Secret Sharing, and an Isolation Forest anomaly detection model.
- **A Web-Based Automation Framework for Instagram Profile Forensics** [IJRDET] (Dec 2025)
  - Developed an automated, non-invasive digital forensic tool using FastAPI and Flutter to streamline evidence acquisition for investigations. Integrated Playwright engineered automated SHA-256 evidence hashing with timestamp validation and built dynamic PDF generation with ReportLab to produce court-admissible, tamper-evident forensic reports.

Skills

- **Cybersecurity & Forensics:** Vulnerability Assessment, Mobile & Digital Forensics, Network Scanning & Analysis.
- **Security & Forensic Tools:** Burp Suite, Wireshark, Nmap, Kali Linux, Metasploit, Recon-ng, Maltego.
- **Programming Languages:** Python, C, C++, Java, Dart, Go, Rust, SQL, JavaScript, HTML, CSS, Bash/Shell Scripting.
- **Backend & Web Technologies:** FastAPI, WebSockets, JWT Authentication, Role-Based Access Control (RBAC).
- **Mobile & App Development:** Flutter, Firebase
- **Distributed Systems:** Blockchain, IPFS, P2P Protocols.
- **AI & Applied Data:** NLP, Semantic Similarity, Text Classification, Scikit-learn/Spacy.
- **Tools:** Docker, Git/GitHub, Linux (Parrot, Kali, Fedora, Ubuntu), Android Studio, VS Code, Jupyter Notebook, LaTeX

- **Soft Skills:** Team Leadership, Problem Solving, Communication, Project Management, Technical Writing & Presentation.
- **Spoken Languages:** English, Hindi, Telugu, Kannada, Marathi, German (Beginner).

Certifications

- **Cyber Crime and Cyber Forensics Course** – National Law University Odisha – **[Scored 90%]** (Aug 2026)
- **Ethical Hacking Awareness Workshop** – XploitForge (July 2025)
- **Machine and Deep Learning Techniques Webinar** – Computer Society of India (CSI) (Mar 2025)
- **Cybersecurity with Ethical Hacking Workshop** – IIT Hyderabad (Jan 2025)
- **Cybersecurity Analyst Job Simulation (IAM)** – Tata Group & Forage (Dec 2024)
- **Ethical Hacking Course -** NPTEL, India – **[Scored 71%]** (Oct 2024)
- **Project School Certification** – Keshav Memorial Engineering College (Feb 2024)
- **NCC A Certificate** – National Cadet Corps (NCC), India (June 2019)

Achievements

- **1st Place Winner - National Level Coding Contest,** Aurora Technical and Research Institute (Jun 2023)
  - Demonstrated exceptional problem-solving skills by achieving a 95% accuracy rate in solving complex coding challenges, outperforming 200 participants.
- **Winner - TECHTONIC '24 Hackathon,** Keshav Memorial Engineering College (May 2024)
  - Led a team to design and implement a cutting-edge solution that enhanced website performance, winning 1st place among 30 teams.

Activities and Societies

- **Social Media Team Head – Computer Society of India (CSI) KMEC**– Spearheaded digital outreach and content strategy, driving student engagement and promoting technical events and workshops across digital platforms.
- **National Service Scheme (NSS)** – Contributed to social initiatives and community service projects.
- **Event Organizer** – College Technical & Cultural Events - Managed and coordinated various college events, ensuring smooth execution and participation.
- **NCC Candidate** – Developed discipline, leadership, and teamwork through National Cadet Corps training.
- **Martial Arts** – Practiced martial arts, enhancing discipline, focus, and self-defence skills.